1.Abstract

The methodology underlying our model development is founded on two fundamental principles of machine learning. The first principle centres on **feature optimization**, which includes both the elimination of non-predictive variables and the generation of novel features through feature engineering. This comprehensive feature optimization framework was instrumental in constructing an optimal set of predictive variables. The second principle leverages **model ensemble** techniques, wherein the integration of two or more distinct models reduces the individual estimation errors, thereby enhancing the composite model's predictive accuracy.

Our methodology offers significant potential for advancing early Alzheimer's Disease and Related Dementias (AD/ADRD) prediction through optimization of prospective data collection protocols. Feature elimination analysis revealed that predictive power is concentrated in a specific subset of variables, enabling more efficient data collection strategies by identifying and excluding non-contributory measurements. Furthermore, feature importance analysis identified a subset of important predictors, for instance those associated with educational parameters, suggesting that expanded data collection in these domains (e.g., detailed educational background including university specializations) could yield improved predictive capabilities.

Additional methodological considerations, including error estimation and the modelling of temporal progression of cognitive scores, which may enhance clinical diagnostic processes and inform treatment strategies, are detailed in Section 5.

2. Methodology

2.1 Data preprocessing

As a first preprocessing step we converted a subset of the original features[1] from ‘categorical’ to ‘numerical’. Furthermore, we replaced the missing values for all the categorical features which were not pre-processed on the first step (e.g. ‘gender’) with an empty string[2].

2.2 Feature elimination

Several of the 184 features provided in the dataset (including ‘year’ and excluding ’uid’) are likely to have no predictive power and are therefore likely to lead to overfitting. To eliminate redundant features, we employed the following methodology:
a) we randomly selected 80% of the patients (i.e. 80% of the unique patient identifiers) who were used as the training set and employed the remaining 20% as the evaluation set).
b) we fit a catboost model (Categorical Boosting) on the train data. We stopped the iterations when the model performance on the evaluation set started deteriorating [3] .
c) We used Shapley Values to assess the importance of each feature for the ‘best’ model selected on step a). Shapley Values are considered the most theoretically sound methodology to assess feature importance (Rozemberczki et al, 2022). On each iteration we removed the 5 features with

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[1]The features were converted: 'age_03', 'edu_gru_03', 'n_living_child_03', 'glob_hlth_03', 'bmi_03','age_12', 'edu_gru_12', 'n_living_child_12', 'glob_hlth_12', 'bmi_12', 'satis_ideal_12', 'memory_12', 'rameduc_m', 'rafeduc_m', 'rrfcntx_m_12', 'rsocact_m_12'

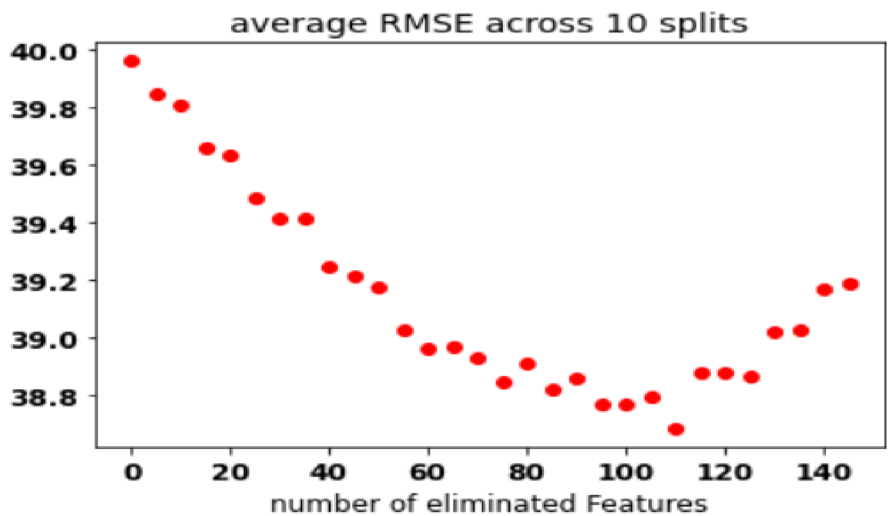
[2] Removing missing values for categorical features is a prerequisite to apply the CatBoost model.

[3] This is commonly referred to as ‘Early Stopping’ in the machine learning literature.

the lowest Shapley Value and iterated the process 30 times[4]. At the end, each feature has either been retained in the final model, thus receiving a score of 30, or it has been eliminated after a certain number of iterations (if a feature is eliminated after 5 iterations then that feature gets a score of 5).

To reduce the noise and make results more robust, we repeated the {train, evaluation} split described in a) and carried out the recursive feature elimination process 10 times. In the end we selected the optimal number of features N as the one giving the lowest average error on the validation set over the 10 repetitions. The feature importance scores (computed as explained in c)) were then averaged across the 10 random splits and the top N features were retained. The average RMSE averaged across the 10 {train, evaluation} splits is displayed in Figure 1. Note that U-shape curve as using all the features leads to overfitting and eliminating too many features leads to a similar decrease of predictive power.

Figure 1. Average RMSE as a function of the number of features eliminated



2.3 Feature engineering

Feature engineering refers to the process of creating new features from the original set and retaining those which can help improve the forecast. By considering the mean, max, count (i.e. number of finite values) and sum of all the features retained as described in Section 2.2, we created 1453 additional features. To assess which of the features generated through our feature engineering process can add value, we first carried out a semi-manual procedure in order to filter out irrelevant features[5]; after reducing the number of candidate features to a more reasonable number (~300), we then used the same approach described in 2.2 and retained only a subset of the original features.

In the end we selected 174 features which were used for the subsequent modelling. Seventy-four were part of the original set of features and 100 were generated through the feature

[4] On iteration 1, for instance, this leads to a decrease of the total number of features from 184 to 179, on iteration 2, only 174 features were retained and so on.
[5] For instance those with a low number of finite values or with a very low correlation with the target variable.

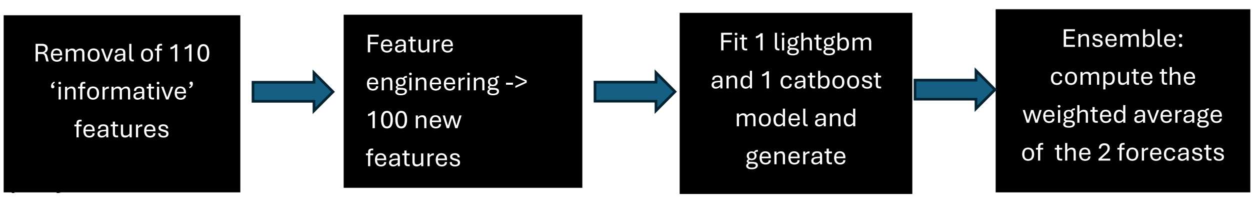
engineering process.

2.4 Choice of the models and ensemble

For the modelling we focused on gradient boosting-based methods given their proven value as showcased in many machine learning competitions. In particular, we fitted one Lightgbm[6] model and one Catboost[7] model on the training set; in both cases we optimized for the learning rate, depth of the trees and number of trees, leaving the other hyperparameters to their default settings. These 3 hyperparameters were selected by splitting the training set repeatedly into {train, evaluation} subsets and by retaining the hyperparameters that provided the lowest errors on the unseen data of the evaluation set[8]. In the end, the test set forecast which resulted in the 5th place in the final leaderboard was obtained as linear combination of the forecasts of the Lightgbm and Catboost models forecast with weights of 25% for the former and 75% for the latter[9].

The entire modeling process is summarized in Figure 2.

Figure 2. Summary of the modelling pipeline



3. Performance Analysis

In this section we elaborate on two topics mentioned in the specification for the 'Performance analysis' Section which we feel deserve special attention.

3.1 Uncertainty estimation

Although this competition required to come up with a 'point estimate' of the composite score (i.e. a single number), assessing the ***uncertainty*** of each individual estimate is also important. To obtain confidence intervals (CI), 'conformal prediction' (CP, Tibshirani, 2023) is a powerful tool developed in recent years which allows us to compute CI for arbitrary confidence levels (e.g. 90%) for any given forecasting model. 'Local conformal prediction', in particular, allows us to estimate subject-specific CI. The estimates computed by our best model on the test set and the corresponding 90% CI, sorted by composite score forecast of the 1105 patients, are shown in Figure 3. On the left we displayed the CI obtained using 'split CP', which produces the same error estimate for all subjects and on the right, we used 'local CP'[10], which generates subject-specific confidence intervals. Having subject-specific error estimates can be very valuable because it may assist clinicians in a better assessment of the individual estimates generated by the model.

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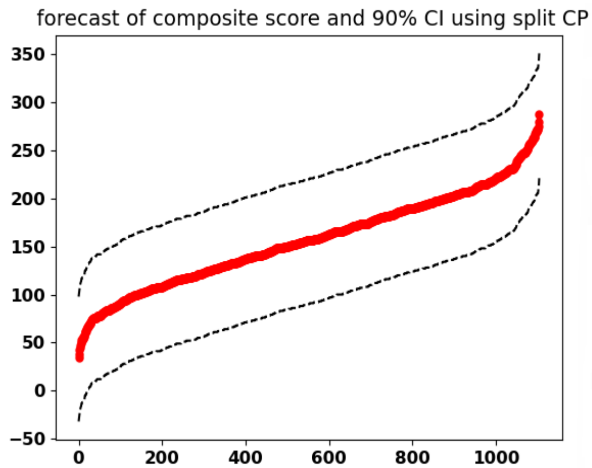
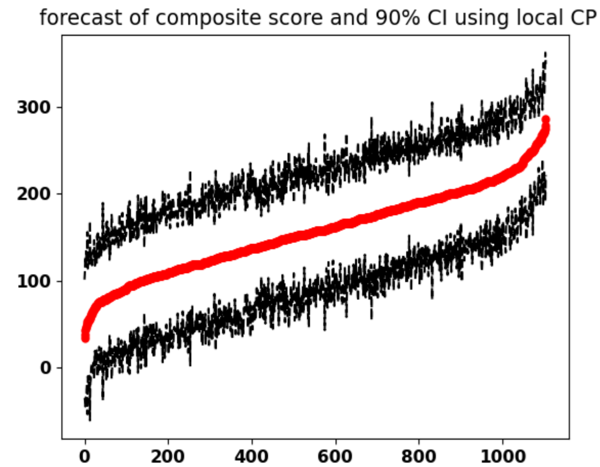
[6] Lightgbm is an open-source Python-based package for gradient boosting developed by Microsoft.
[7] Catboost is an open-source Python-based package for gradient boosting developed by Yandex.
[8] The {learning rate, depth of tree, number of trees} selected were {0.05,5, 250} for the Lightgbm model and {0.05,5, 800} for the Catboost model.
[9] The weights of the two models were optimized using the same {train, evaluation} split approach on the training data.

Figure 3. Forecast of the composite score and of its 90% confidence interval obtained using 'split' conformal prediction (left, same error for all subjects) ad 'local' conformal prediction where the error estimate is subject-dependent (right).





2.5 Feature importance

Feature importance analysis, which tries to assess which are the most important drivers of the prediction of a given fitted model, is of paramount importance because it helps increase explainability and reduce the perception of the model as a 'black box'. Several methods have been developed to estimate feature importance; we focus here on the use of Shapley values that is considered the 'gold standard' for this task (Rozemberczki, 2022).

Figure 4 displays the 10 most important drivers of the prediction for the Catboost model. As expected, the age at the time of assessment ('age_12') and the 'year of assessment' appear in the list; however, and perhaps quite surprisingly, 2 variables related to education ('edu_gru_12') and mental activity ('reads_12') appear in the top 3 features with 2 other features related to social and family dynamics[11] also making the list. On the other end, the Body Mass Index is the only feature related to physical or mental health which came up as important driver in the prediction whereas we expected this type of variables to be highly predictive.

4. Bias and Ethics Analysis

Bias analysis tries to assess whether a given model tends to be systematically better or worse depending on variables such as gender, age, years of education (Dorne, 2024, Xu, 2022)[12]. Note that we are not trying to assess whether the predictions of our model are different as a function of the selected PA but whether the **absolute errors** are statistically different.

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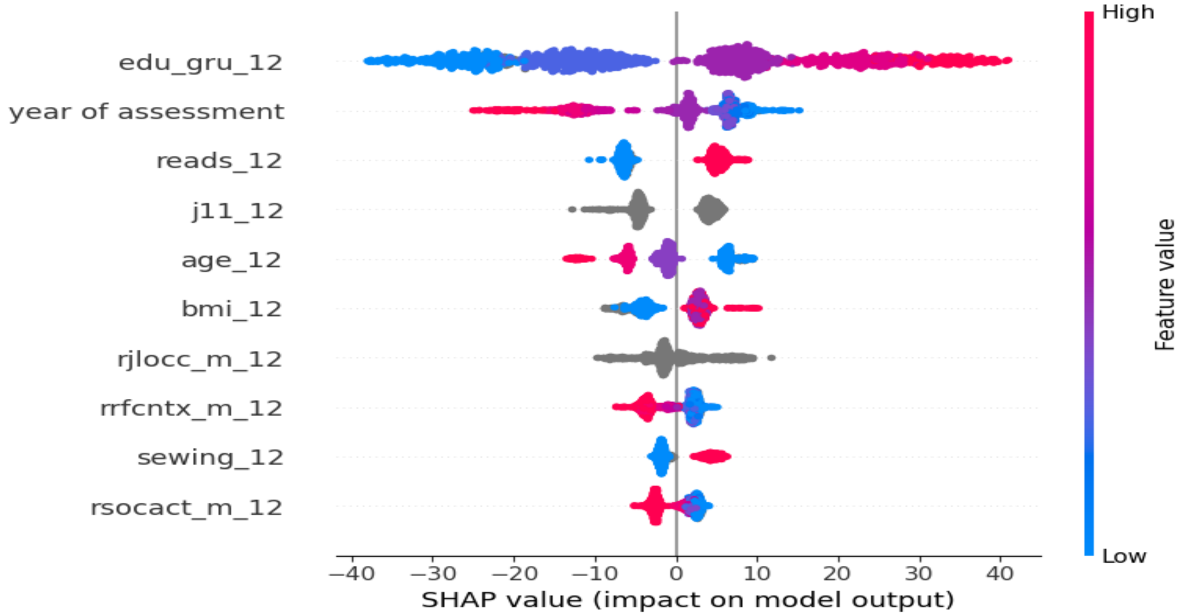
[11] How often they see friends and relatives' and 'How often they have social activities'.

[12] Following the literature we will refer to these variables as 'protected attributes', PA.

Figure 4. 10 most important features according to their Shapley Values.



For the analysis presented in this report we selected a) gender b) binned education level and c) employment status as the attributes to assess. However, the approach described below could be used for any other discrete[13] variable. For each of the selected PAs, we run the Kruskal-Wallis test[14] to assess whether the mean absolute error was different across various categories of the protected attribute (i.e. the null hypothesis is $u_1=u_2= \ \ u_n$, if the PA has *n* categories).

The p-values for the PAs tested (‘ragender': 0.14 ,'edu_gru_03': 0.16, 'edu_gru_12': 0.26, 'employment_03': 0.58, 'employment_12': 0.65} indicate that, for all of them, we cannot reject the null hypothesis at the conventional 5% significance level and therefore that our model is ‘unbiased’ with respect to all the selected protected variables.

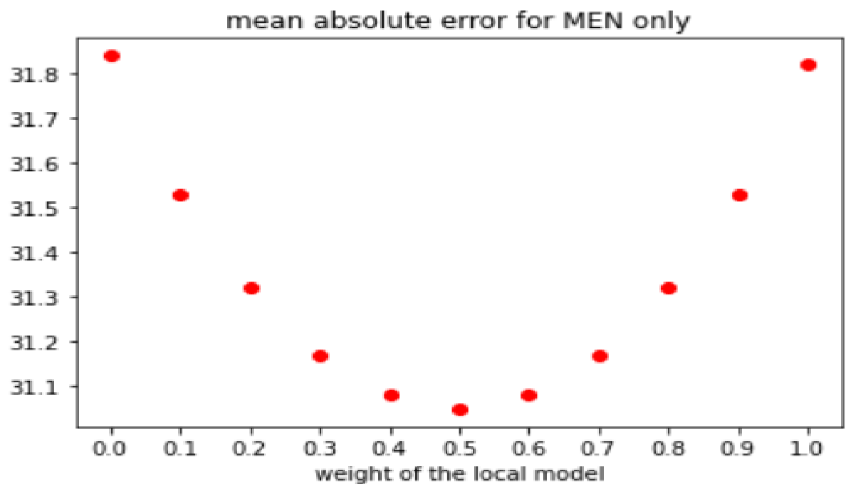
4.1. Bias mitigation

Assume, however, that the Kruskal-Wallis test indicated that the model is biased with respect to one or more of the PAs. To address this, we focus specifically on the gender variable as an example, because its lowest p-value suggests a possible issue with the model. The estimates for males (31.83) are characterized by a higher mean absolute error than those for women (29.06). We note that in our gender example, the higher error for men stems from the fact that only 41.8% of the subject in the training set are male and therefore, the number of datapoints for men is lower than women[15]. This suggests that categories underrepresented in the training set (e.g. ethnic minorities) may end up with worse estimates. To mitigate this, we propose to train a model using

[13] Had any of the PA been continuous we would have fitted the absolute error on the variable of interest using a linear regression with intercept on the PA values and assessed the statistical significance of the estimated slope coefficient.
[14] We used the Kruskal-Wallis test and not a more standard ANOVA as the analysis of the errors showed that they do not follow a Gaussian distribution.
[15]This can be proved by randomly sampling a number of women equal to the number of men in the original training set in order to train a model on a balanced dataset and then assessing the error on unseen data.

only data from men which we refer to as ‘local model’ (LM). Its strength lies in the fact that the LM is able to capture men-specific patterns at the cost of an increased risk of overfitting due to the use of a smaller training dataset. The ‘global model’ (GM) trained on all data has exactly the opposite strengths and weaknesses. The plot below shows that the linear combination {w* LM forecast +(1-w)* GM forecast} of the GM and protected variable-specific LM is able to substantially reduce the estimation error by exploiting the specific strengths of both models. The idea just presented is general and can be applied to any discrete PA; however, care should be taken to fine-tune the hyperparameters of the LM to account for the reduced number of training data.

Figure 5: Mean Absolute error for the gender protected variable obtained by linearly combining a global model and local variable-specific model.



5. Future directions

The clinical implications of an early diagnosis of AD/ADRD are substantial and extend well beyond the scope of this report. Here, we focus on some methodological issues which can lead to improved estimates of the composite score and result in a more accurate early diagnosis.

5.1 More data->better models

The importance of ‘data’ in the accuracy of the forecast of any machine learning model cannot be understated. The collection of additional data and/or the use of other datasets not provided in the competition can be a key factor in improving any model. In this respect, the option of generating ‘synthetic’ data using Generative Adversarial Networks (GANs, see for instance Goodfellow,2014 and Aggarwal,2021), which are widely used in both finance and computer vision, is worth exploring. We propose to generate synthetic data, train our model using a combination of real data and synthetic data, and assess the accuracy of the model on unseen real data; we believe this is a promising avenue to consider.

5.2 Optimization of feature collection

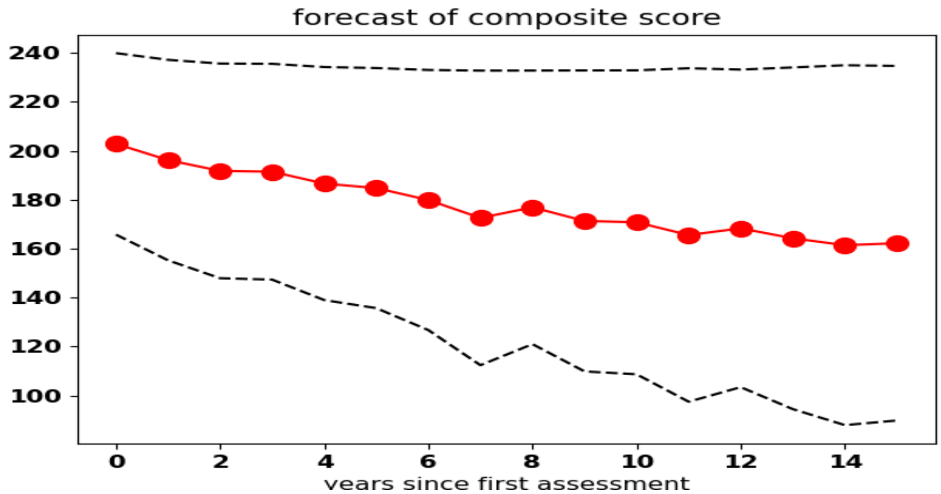
The feature elimination process described in Section 2.2. showed that only a subset of the features provided in the competition has real predictive power, which will enable more efficient data collection strategies by excluding non-contributory measurements from the collection. On the other hand, our feature importance analysis showed that certain features are strong predictors (e.g. features related to 'education'); this suggests that collecting additional data related to those features can enhance the accuracy of the forecast. We could, for instance, gather information on the 'subject studied in university' or 'education level of siblings', as 'education' has a high predictive power. Extending the data collection process to features likely to be informative is expected to improve the accuracy of the composite score estimate.

5.3 Cognitive decline estimation

Generating forecasts for 1,2,3, N years in the future is valuable as it can be used to assess the cognitive decline of each subject. This idea is summarized in Figure 6, which shows the progression of the composite score as a function of the years in the future for a hypothetical patient[16], generated at the time of the first assessment (year 0), together with its 90% confidence interval. To create a model which can estimate the evolution of the composite score with time, we need to include in the training set only subjects for whom at least 2 assessments are available[17] and use the time elapsed between assessment 1 and 2 as an additional feature (the '*time'* feature). The key idea is to use the difference in scores between assessments 2 and 1 as the target variable, and not the composite score itself. Then, to generate the prediction for a patient assessed today, we input their features into the model and generate the point estimate for 1 year ahead by setting the 'time' feature to 1, for 2 years ahead by setting 'time' to 2, and for N years ahead by setting 'time' to N. Confidence intervals can be computed using conformal prediction as explained in Section 3.1. To generate the total composite score forecast we obviously need one 'base' model (like the one created for this competition) which produces the absolute forecast of the score S, as the 'delta' model described in this section only generates estimates of the difference D between scores, so that the final forecast is S+D.

We believe that the combination of error estimate and progression of the score with time can offer additional information with respect to a single point estimate and could assist the clinician in the diagnosis and ultimately lead to a better treatment.

Figure 6. Estimate of the composite score progression as function of the years in the future and corresponding 90% confidence interval



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[16] In the plot we have represented the decline of the composite score with time (i.e. the older the subject gets, the lower the expected score), the noise of the estimates and the higher standard deviation for longer time horizons

[17] Generalizing this to patients with more than two assessments is trivial. If a patient has n assessments, we can use all the [n (n-1)/2] distinct couples of assessments in the training set.

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